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Statistics and frequency-domain moveout for multiple-taper receiver functions

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SUMMARY

The multiple-taper correlation (MTC) algorithm for the estimation of teleseismic receiver functions (RFs) has desirable statistical properties. This paper presents several adaptations to the MTC algorithm that exploit its frequency-domain uncertainty estimates to generate stable RFs that include moveout corrections for deeper interfaces. Narrow-band frequency averaging implicit in spectral cross-correlation restricts the MTC-based RF estimates to resolve *Ps* converted phases only at short delay times, appropriate to the upper 100 km of Earth's lithosphere. The *Ps* conversions from deeper interfaces can be reconstructed by the MTC algorithm in two ways. Event cross-correlation computes a cross-correlation of single-taper spectrum estimates for a cluster of events rather than for a set of eigenspectrum estimates of a single *P* coda. To extend the reach of the algorithm, pre-stack moveout corrections in the frequency domain preserves the formal uncertainties of the RF estimates, which are used to weight RF stacks. Moving-window migration retains the multiple-taper approach, but cross-correlates the *P*-polarized motion with time-delayed *SH* and *SV* motion to focus on a *Ps* phase of interest. The frequency-domain uncertainties of bin-averaged RFs do not translate directly into the time domain. A jackknife over data records in each bin stack offers uncertainty estimates in the time domain while preserving uncertainty weighting in the frequency-domain RF stack.

Key words: Time-series analysis; Fourier analysis; Probability distributions; Body waves; Coda waves; Wave scattering and diffraction.

1 INTRODUCTION

Teleseismic receiver functions (RFs) constitute a method for estimating the mode-converted scattered waves associated with the main *P* and *S* body phases from earthquakes (Fig. 1). Phinney (1964), Vinnik (1977), Berkhout (1977) and Langston (1981) developed a deconvolution procedure for first-arriving *P* waves, which are now known as RFs for their dependence on layered structure directly beneath the station. RFs have been used to probe for converted waves associated with Moho depth variations (Zhu & Kanamori 2000; Mele & Sandvol 2003; Bannister *et al.* 2004; Gilbert & Sheehan 2004; Leahy & Park 2005; Lodge & Helffrich 2006; Prodehl *et al.* 2013; Levin *et al.* 2016), subducting lithosphere both active and fossil (Bostock 1998; Kosarev *et al.* 1999; Park *et al.* 2004; Shiomi *et al.* 2004; Bannister *et al.* 2007; Bianchi *et al.* 2010), transition zone interfaces (Chevrot *et al.* 1999; Lebedev *et al.* 2002; Lawrence & Shearer 2006; Huckfeldt *et al.* 2013; Liu & Pavlis 2013; Liu *et al.* 2016) and anisotropic layers (Kosarev *et al.* 1984; Levin & Park 1997, 2000; Bostock 1998; Leidig & Zandt 2003; Savage *et al.* 2007; Liu *et al.* 2015). Backazimuth stacking for anisotropic effects is discussed by our companion paper Park & Levin (2016). Although mainly applied to first-arriving compressional waves (*P* and the *PKP* family of phases), shear-wave RFs have also been studied to examine *S*-to-*P* scattering from mantle interfaces (Farra & Vinnik 2000; Rychert *et al.* 2005; Ford *et al.* 2010).

RF estimation is subject to several sources of numerical instability and uncertainty. The earliest method for deconvolution, spectral division, is vulnerable to the stochastic near-zero values of the denominator. Regularization by means of a 'water-level' constant in the denominator is popular (e.g. Ammon 1991) but the analyst risks damping the low-amplitude, high-frequency signal (Park & Levin 2000). Deconvolution in the time domain (Abers *et al.* 1995; Gurrola *et al.* 1995; Ligorria & Ammon 1999; Poppeliers & Pavlis 2003a,b; Escalante *et al.* 2007) avoids the spurious narrow-band oscillations of spectral division, but, if performed on raw seismic time-series with a least-squares constraint, the estimated RF will be dominated by the largest-amplitude frequency range in the seismic phases (Park & Levin 2000).

Park & Levin (2000) introduced the multiple-taper correlation (MTC) RF estimator, which operates in the frequency domain. Spectral correlation is more stable than spectral division, and offers an objective estimate of uncertainty for the RF in the frequency domain, based on the spectral coherence between the vertical and the two horizontal components. Using this uncertainty estimate as a

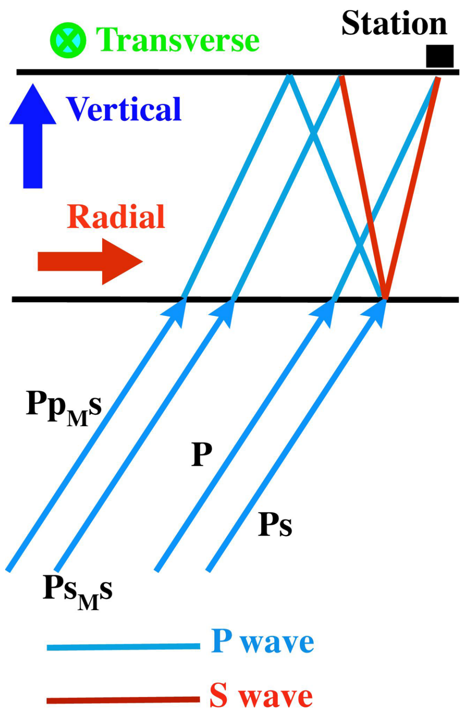

Figure 1. Schematic diagram of ray paths used in the interpretation of receiver-function estimates.

weighting parameter when stacking data from different events, the MTC-based RF estimates are found without free parameters. Using this coherence measure, Park & Levin (2000) showed that much of the spectral energy of horizontal motion is in fact uncorrelated with spectral energy on the vertical component. This lack of correlation suggests that the largest source of noise in RF estimation is signal-generated, partly associated with a poorly correlated scattered wavefield that the incoming body wave generates as it strikes Earth's rough topography and near-surface structure (Vernon *et al.* 1991). The MTC RF estimates have found wide use in studies of structure shallower than 100 km, in particular the retrieval of signals from the transverse horizontal component (Levin & Park 2000; Park *et al.* 2004; Liu *et al.* 2015).

The restriction of MTC RFs to shallow lithospheric structure is caused by an implicit frequency average associated with spectral correlation. A spectrum estimate from a simple untapered Fourier transform involves an average of spectrum estimation over a bandwidth twice the Rayleigh frequency $f_R = 1/T$, where T is the duration of the analysed time-series. In principle, spectral division can reconstruct scattered-wave pulses that are delayed (or advanced) relative to the P wave by a time $\tau \sim T/2$. Park & Levin (2000) compute RFs using three mutually orthogonal Slepian data tapers with time-bandwidth product $p = 2.5$, so that their spectral correlation averages over an interval $5f_R$. As a result, the deconvolution afforded by the MTC RFs is limited nominally to delay times $\tau \sim T/5$, for example, $\tau \approx 16$ s for a 80-s record of P coda. In practice, the MTC RFs suffer amplitude loss in the time domain at $\tau > T/5$. (Further comments on this topic can be found in the Supporting Information.)

This paper presents methods that we and others have developed to extend MTC RF estimates in time, while also preserving their desirable statistical behaviour. The two methods proposed, multiple-event correlation and time-delayed correlation, can be shown to give consistent predictions for P-to-S conversions at long delay time, which encourages confidence in their use. Accurate resolution of RFs at longer delay times τ requires the introduction of moveout corrections in the RF estimator. RF migration has typically been applied in the time domain (Sheehan *et al.* 2000; Morozov & Dueker 2003a,b; Poppeliers & Pavlis 2003a,b). This approach has been generalized by Chen *et al.* (2005a,b), Shang *et al.* (2012), Liu & Levander (2013) and Zhang & Fredricksen (2015) to amplitudes of scattered waves from 3-D structures. Time-domain stacking discards the frequency-dependent uncertainties that the MTC algorithm provides. Good and marginal RF estimates carry equal weight in a time-domain stack. In this paper we describe several moveout algorithms that preserve the coherence-derived RF uncertainties, and therefore routinely downweight noisier RFs in a stacked estimate.

The sections of the paper address a number of algorithmic topics. Section 1 outlines the MTC algorithm, and modifies the original technique to cross-correlate the ray-based L, Q and T components, which correspond roughly to the P, SV and SH polarizations. Section 2 presents *a posteriori* tests of the statistical assumptions of the MTC RF estimates, and also formulates an alternate 3-component cross-correlation using the singular-value decomposition (SVD). Section 3 develops a frequency-domain migration algorithm for MTC RFs. Section 4 describes an event-stacking algorithm for computing RFs at long delay times. Section 5 develops a moving-window algorithm for retrieving MTC RFs at long delay times (Helffrich 2006; Abt *et al.* 2010) that weighs the stack by RF uncertainties in the frequency domain. Section 6 presents a jackknife method for estimating time-domain uncertainties of the stacked RFs. Section 7 summarizes our results.

2 MULTIPLE-TAPER CORRELATION RECEIVER FUNCTIONS

Following Park & Levin (2000), we start with three N-point time-series $\{u_n^R, u_n^T, u_n^Z\}_{n=0}^{N-1}$, with sampling interval Δt and duration $T = N\Delta t$, representing radial, transverse, and vertical seismic motion. We assume that the ground-motion components are at least partially stochastic and have spectra $\mathcal{U}_Z(f)$, $\mathcal{U}_R(f)$ and $\mathcal{U}_T(f)$, which we can estimate with tapered Fourier transforms of the three time-series. If we knew the ground-motion spectra exactly, RFs could be computed in the frequency domain via the division of the horizontal spectra $\mathcal{U}_R(f)$ and $\mathcal{U}_T(f)$ by the vertical spectrum $\mathcal{U}_Z(f)$. The discrete Fourier transform (DFT), however, cannot retrieve the spectrum exactly for stochastic time-series, and taking the ratio of two DFTs is numerically unstable.

Rather than dividing two spectrum estimates at a frequency point f, Park & Levin (2000) estimate the spectral ratio from the correlation of spectra averaged over a narrow frequency band $(f - f_w, f + f_w)$. The statistical properties of the DFT make correlation the stabler choice. The output of a DFT of a time-series, multiplied by a data taper $\{w_n\}_{n=0}^{N-1}$, is the frequency-domain convolution of the 'true' spectra $\mathcal{U}_Z(f)$, etc., with the DFT of the taper (Thomson 1982; Park *et al.* 1987; Prieto *et al.* 2009). A taper whose DFT is strongly concentrated in a narrow bandwidth around $f = 0$ can provide a reliable local average of spectral properties in the neighbourhood of a frequency f. The Slepian data tapers $\{w_n^{(k)}\}_{n=0}^{N-1}$, for $k = 0, 1, \dots K - 1$, are optimized for this local average within a user-chosen frequency band (Thomson 1982), and can be computed from the eigenvectors of a Toeplitz matrix (Lees & Park 1995) or of a tridiagonal matrix, as described by Percival & Walden (1993), pp. 386–387. We choose the frequency band to have half-bandwidth $f_w = pf_R$, where the Rayleigh frequency $f_R = 1/T$ is the nominal frequency resolution of the DFT, and $p \geq 1$ is called the time-bandwidth product. The first $2p-1$ Slepian tapers have good concentration within the frequency band $(f-f_w, f+f_w)$ when averaging spectral properties (Park *et al.* 1987).

Using $K=3$ Slepian data tapers computed for time–bandwidth product $p=2.5$, we compute estimates $Y^{(k)}(f)$ of the spectrum with the DFT, for example,

$$Y_R^{(k)}(f) = \sum_{n=0}^{N-1} u_n^R w_n^{(k)} \exp(i 2\pi n \Delta t). \tag{1}$$

The spectrum estimates $Y_R^{(k)}(f)$ are called 'eigenspectra'. The Slepian tapers are eigenvectors of a symmetric matrix, and therefore mutually orthogonal. Taper orthogonality implies that the $K=3$ eigenspectra of a stochastic time-series computed at any particular frequency f are statistically independent. Multiple-taper RFs $H_R(f)$ and $H_T(f)$ are defined as a spectral correlation of the radial and transverse spectrum estimates $Y_R^{(k)}(f)$ and $Y_T^{(k)}(f)$, respectively, with the vertical spectrum estimates $Y_Z^{(k)}(f)$ over the $K=3$ eigenspectra, for example,

$$H_R(f) = \left(\frac{\sum_{k=0}^{K-1} \left(Y_Z^{(k)}(f)\right)^* Y_R^{(k)}(f)}{\left(\sum_{k=0}^{K-1} \left((Y_Z^{(k)}(f))^* Y_Z^{(k)}(f) + S_o(f)\right)\right)} \right) \tag{2}$$

where the asterisk denotes complex conjugation and $S_o(f)$ is the squared-spectrum estimate of the pre-event vertical motion at frequency f. The pre-event spectrum estimate functions as a frequency-dependent water-level damping parameter.

We express formal uncertainty in $H_R(f)$ and $H_T(f)$ with a formula based on spectral coherence, for example,

$$\text{variance}\,(H_R(f)) = \frac{1 - C_{ZR}^2(f)}{(K-1)C_{ZR}^2(f)} |H_R(f)|^2 \tag{3}$$

where the coherence $C_{ZR}(f)$ is defined by

$$C_{ZR}(f) = \frac{\sum_{k=0}^{K-1} \left(Y_Z^{(k)}(f)\right)^* Y_R^{(k)}(f)}{\left(\sum_{k=0}^{K-1} \left(Y_R^{(k)}(f)\right)^* Y_R^{(k)}(f)\right)^{\frac{1}{2}} \left(\sum_{k=0}^{K-1} \left(Y_Z^{(k)}(f)\right)^* Y_Z^{(k)}(f)\right)^{\frac{1}{2}}}. \tag{4}$$

Examples of these computations (Park & Levin 2000) show that the formal uncertainty of $H_R(f)$ and $H_T(f)$ varies strongly with f, and among P coda from different events. The rectilinear motion of the direct P leads to a strong baseline correlation between the vertical and radial ground-motion time-series $\mathbf{u}^Z$ and $\mathbf{u}^R$, so that the phase of $H_R(f)$ varies slowly compared to the phase of $H_T(f)$.

Background seismic noise is composed of myriad travelling waves excited by ocean waves, wind stress and other sources (Shapiro & Campillo 2004), and the P coda are dense with signal-generated side-scattered and back-scattered waves (Vernon *et al.* 1991). As a result, the uncertainties of the frequency-domain RFs $H_R(f)$ and $H_T(f)$ should be uncorrelated for distinct seismic events. To take full advantage of the frequency dependence of the formal variance estimates, RF stacks should be performed in the frequency domain, with a weighted average from a standard regression formula (Park & Levin 2000). Stacking can be performed over equal intervals of epicentral distance (Fig. 2), to check for the moveout of converted phases, or of backazimuth (Fig. 3), which can highlight polarity changes associated with anisotropic layers or dipping interfaces (Park & Levin 2016).

The time-domain radial and transverse RFs $H_R(\tau)$ and $H_T(\tau)$ are the inverse Fourier transforms of $H_R(f)$ and $H_T(f)$. If the variances of $H_R(f)$ and $H_T(f)$ were uncorrelated at different frequencies, the uncertainty for the time-domain RFs would be time-invariant. However, the background seismic noise is not white, but highly

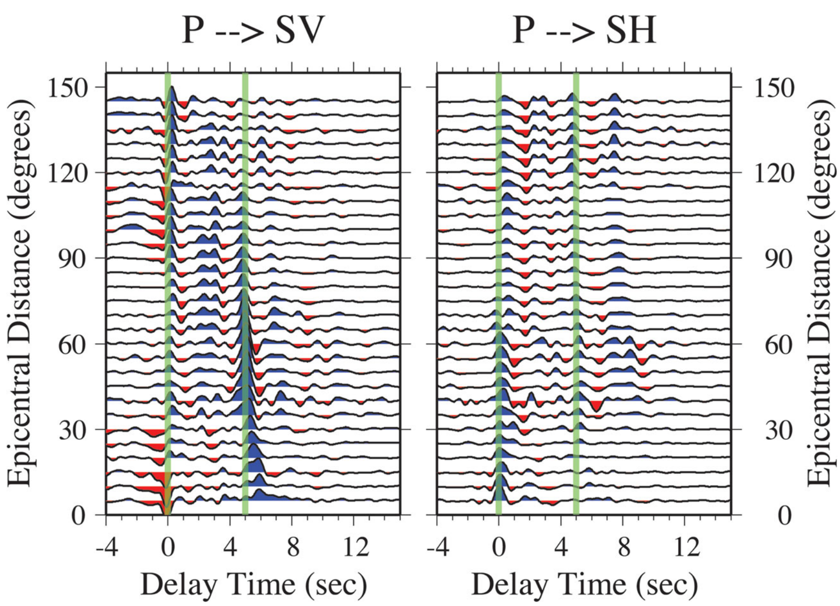

Figure 2. Multiple-taper correlation (MTC) receiver functions for station RAYN of the Global Seismographic Network (GSN), computed for 60-s time windows with frequency cut-off $f_c = 2.0$ Hz as a function of epicentral distance Δ. We filter the RFs to avoid bandedge ringing via a squared-cosine taper in the frequency domain, with half-amplitude at 1.0 Hz. We use seismic events in the 50°–110° backazimuth sector, with frequency-domain RFs averaged in Δ-bins of 5° half-width, spaced by 5°. The RFs are referenced to a ray-based coordinate system (LQT) rather than epicentral coordinates (vertical-radial-transverse), and therefore lack a large positive pulse at zero time delay that arises from the horizontal component of the P wave. The vertical line at ∼5-s delay identifies the Moho-converted Ps phase, which suffers moveout to larger time delay as Δ decreases.

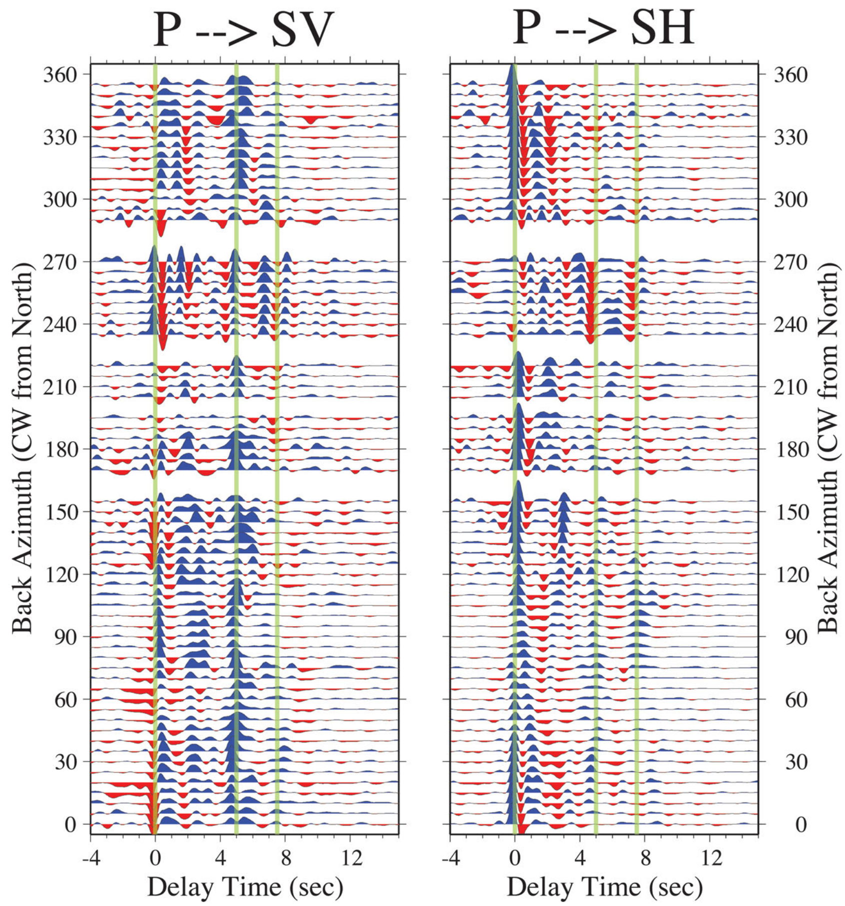

Figure 3. Multiple-taper correlation (MTC) receiver functions for station RAYN of the Global Seismographic Network (GSN), computed with frequency cut-off $f_c = 2.0$ Hz with a 60-s time window. We filter the RFs to avoid bandedge ringing via a squared-cosine taper in the frequency domain, with half-amplitude at 1.0 Hz. We use 720 seismic events with Richter magnitude $M > 6.0$ in 1997–2005. We averaged frequency-domain RFs in backazimuth bins of 5° half-width, spaced by 5°. The RFs are referenced to a ray-based coordinate system (*LQT*) rather than epicentral coordinates (vertical-radial-transverse), and therefore lack a large positive pulse at zero time delay that arises from the horizontal component of the *P* wave. The vertical line at ∼5-s delay identifies the Moho-converted *Ps* phase. Levin & Park (2000) identified the *Ps* phase at 7–8-s time delay as arising from an anisotropic shear zone at the upper-mantle Hales discontinuity, that is, P_{Hs}. A tilted north–south axis of symmetry associated with anisotropy near the Hales discontinuity is indicated by the polarity switch of transverse component of P_{Hs}.

correlated across frequencies for a single seismic event. Therefore the uncertainties in $H_R(\tau)$ and $H_T(\tau)$ are not constant with time. We will return to this issue in Section 7.

Many authors (e.g. Vinnik 1977; Bostock 1998) advocate a rotation of particle-motion coordinates from vertical-radial-transverse to *P*-*SV*-*SH* before RF analysis, in order to visualize better the *P*-to-*SV* conversion at short delay times. (This coordinate system is commonly designated '*LQT*' coordinates.) The *P* and *SV* particle motion components are only approximately orthogonal, because *P* and *SV* incidence angles differ for nonzero slowness *p*. We define the '*L*' component of *LQT* to align with *P* polarization, so that its perpendicular *Q* aligns approximately with the *SV* polarization of the *Ps* converted wave.

In practice, one must choose whether to predict the *P* incidence angle from an empirical or model-based formula, or estimate the vertical-radial ratio from the *P* wave itself. Testing data from several permanent seismological stations, we found that simple cross-correlation of vertical and radial particle motion in the time domain yielded considerable scatter in the apparent incidence angle θ as a function of epicentral distance Δ (Fig. 4). Cross-correlation of low-passed *P*-wave data reduces the scatter of $\theta(\Delta)$ in large-amplitude phases, but low-amplitude signals and high-frequency *PKP* phases remain a concern. By the same token, no single theoretical relation for $\theta(\Delta)$ fits the trend of cross-correlation estimates at seismic stations in distinct tectonic environments, for example, stable cratons versus tectonically active areas. For station RAYN (Ar Rayn, Saudi Arabia) of the Global Seismographic Network (GSN) the empirical formula

$$\theta(\Delta) = \begin{cases} 42 + (20 - \Delta)/3 & \text{for} \quad \Delta \le 80^\circ \\ 16 + (110 - \Delta)^2/150 & \text{for} \quad 80^\circ < \Delta < 118^\circ \\ 9 & \text{for} \quad \Delta \ge 118^\circ \end{cases}$$

works adequately with θ in degrees.

Rather than develop an empirical formula for each station, we recommend using slowness estimates $p(\Delta)$ from earth model IASPEI91 (Crotwell *et al.* 1999) normalized by a fixed crustal velocity that can be varied for different stations. For station RAYN we approximated $\theta(\Delta) = \sin^{-1}(\alpha p(\Delta))$, with compressional velocity $\alpha = 7.5$ km s^{-1} (Fig. 4). A perfect LQT rotation is not critical, though a near-zero $t = 0$ radial-RF pulse helps reveal *Ps* conversions from very shallow interfaces.

The LQT rotation replaces $H_R(t)$ with $H_Q(t)$, which we identify as $H_{SV}(t)$. The '*SV*' RF traces lack the large positive pulse at zero

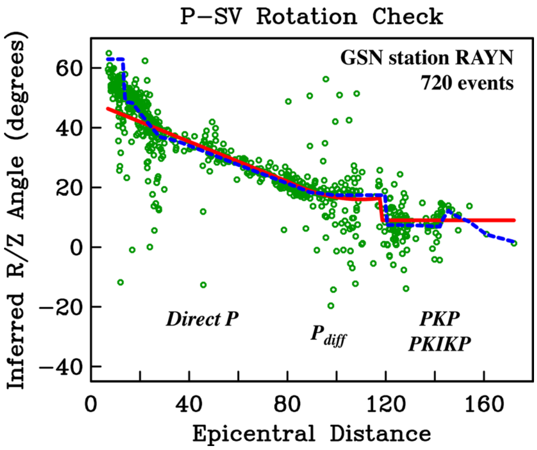

Figure 4. Red solid line plots an empirical formula to define angle θ used in rotation of vertical (*Z*) and radial (*R*) seismic-motion components to ray-based *L* and *Q* components, which approximate the *P* and *SV* polarizations. Blue dashed line plots the radial-vertical angle θ computed from the ray-slowness for the IASPEI91 earth model using TauP software (Crotwell *et al.* 1999). We use $\alpha = 7.5\ \mathrm{km\,s^{-1}}$ at Earth's surface to convert slowness to incidence angle. Green circles show the angle θ between *Z* and *R* for the first 60-s of *P*-coda, inferred from the maximal eigenvector of the matrix formed from time-domain cross-correlation and autocorrelations of $\mathbf{u}^Z$ and $\mathbf{u}^R$. Scatter in θ at RAYN is large for waves that graze the core–mantle boundary, for example, P_{diff}. Station RAYN is typical of stations we have examined in stable continents. Estimated θ for stations in tectonically active regions typically exhibit steeper *P*-wave incidence (smaller average θ) with greater variance about mean values.

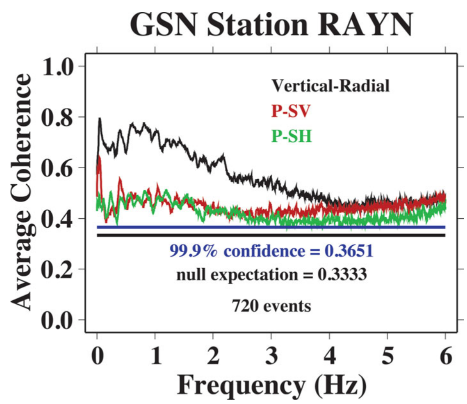

Figure 5. Average estimated coherence $\mathcal{C}(f)$ between components of *P*-wave particle motion, computed for 720 events from 1997–2005 with a 60-s time window. We estimate coherence using three statistically independent eigenspectral estimates computed with three 2.5π-prolate Slepian tapers. For the null hypothesis of white noise, the confidence limits of individual $\mathcal{C}(f)$ estimates can be computed from the *F*-distribution with 2 and 4 degrees of freedom. Combining spectrum estimates from 720 events, the confidence limits of average $\mathcal{C}(f)$ for non-randomness can be computed from the *F*-distribution with 1440 and 2880 degrees of freedom. For this choice, the 99.9 per cent confidence limit for non-randomness occurs at $\mathcal{C} \approx 0.365$, a small amount above the white-noise expected value $\mathcal{C} = 0.333$. Note that average *P-SH* coherence is close to average *P-SV* coherence, suggesting that the larger values of $\mathcal{C}$ between vertical and radial are due largely to the horizontal component of the *P*. Therefore, *P-SV* and *P-SH* scattering have comparable magnitude.

time-delay that characterizes a typical 'radial' RF trace (Figs 2 and 3). In LQT coordinates, the coherence between *P* and *SV* time-series is comparable to the coherence between *P* and *SH* (Fig. 5). This equivalence suggests that *P-SH* scattering is comparable to *P-SV* scattering, and therefore the importance of anisotropy, and perhaps tilted interfaces, in the generation of the *P* coda at RAYN, relative to simple jumps in isotropic seismic velocities on horizontal interfaces (Levin & Park 2000; Park & Levin 2016).

3 STATISTICAL BEHAVIOUR OF STACKED RECEIVER FUNCTIONS

Park & Levin (2000) proposed that MTC RFs *H*(*f*) from individual *P*-wave records be summed using a relative weighting that is inversely proportional to their variances (Fig. 6). We here employ the *a posteriori* χ^2-distribution to test for self-consistency of MTC's statistical assumptions. We employed a data set of 720 earthquakes recorded at 20 samples $\mathrm{s^{-1}}$ at GSN station RAYN by its Geotech KS-54000 borehole sensor during 1997–2005. Technical issues interrupted dataflow at RAYN in 2005, resuming after equipment changes in late 2008. We report tests using data from this single sensor package, which had good signal-to-noise levels while operating. Previous studies of seismic data from station RAYN have characterized its crust and mantle structure as 1-D and fairly simple (e.g. Levin & Park 2000). The earthquake distribution for RAYN has good backazimuth coverage and ranges widely over epicentral distances. Therefore, the RAYN data set forms a good testbed for technique assessment. We confirmed the statistical behaviour at RAYN using data from other stations.

We performed two consistency tests on the statistics of multitaper correlation RF estimation. Basic data-fitting statistics typically assume that data-fit residuals behave as uncorrelated stochastic random variables that follow a Gaussian distribution. The goodness-of-fit metric for the sum of squared residuals, that is, the misfit variance, is the χ^2 distribution. Because we stack the RF at many distinct frequencies, we have many realizations of our data-fitting scheme, and therefore can examine the statistics of our algorithm against both the expectation value and the variance of the χ^2 distribution. We also perform the RF stack in many narrow bins of epicentral distance and backazimuth, with varying numbers of events, so that a range of degrees-of-freedom can be tested.

Using RAYN P-coda data, the residual scatter of MTC RFs has a median value that matches the expectation value of the χ^2 distribution. The uncertainties assigned to the *H*(*f*) are therefore neither too large nor too small. The ensemble of misfit variances has greater variance than predicted by the χ^2 distribution with degrees-of-freedom equal to the number of data. The greater scatter among misfit variances implies that the estimated uncertainties of the single-event *H*(*f*) estimates are the correct size, but are not statistically independent of each other. Correlation among these residuals is not surprising, as we explain below.

If the variance-weighted estimates of RFs have Gaussian statistics and are uncorrelated, the mod-squared misfit variance of sample estimates *H*(*f*) about the sample mean $\overline{H}(f)$, that is,

$$\mathcal{S}^2 = \sum_{m=1}^{M} \frac{|H_m(f) - \overline{H}(f)|^2}{\mathrm{var}\{H_m(f)\}}, \tag{5}$$

should follow a χ^2 distribution with $N = 2M - 2$ degrees of freedom, where *M* is the number of seismic events in the weighted sum. The expectation value $\langle \mathcal{S}^2 \rangle = N$. Earthquakes are distributed unevenly over Earth, so bin-averaging residuals for the RFs graphed in Figs 2 and 3 can be examined for both large and small sample sizes. We compute distinct values of $\mathcal{S}^2$ for each frequency of the DFT up to either 4 Hz or 2 Hz, spacing the frequencies by the nominal

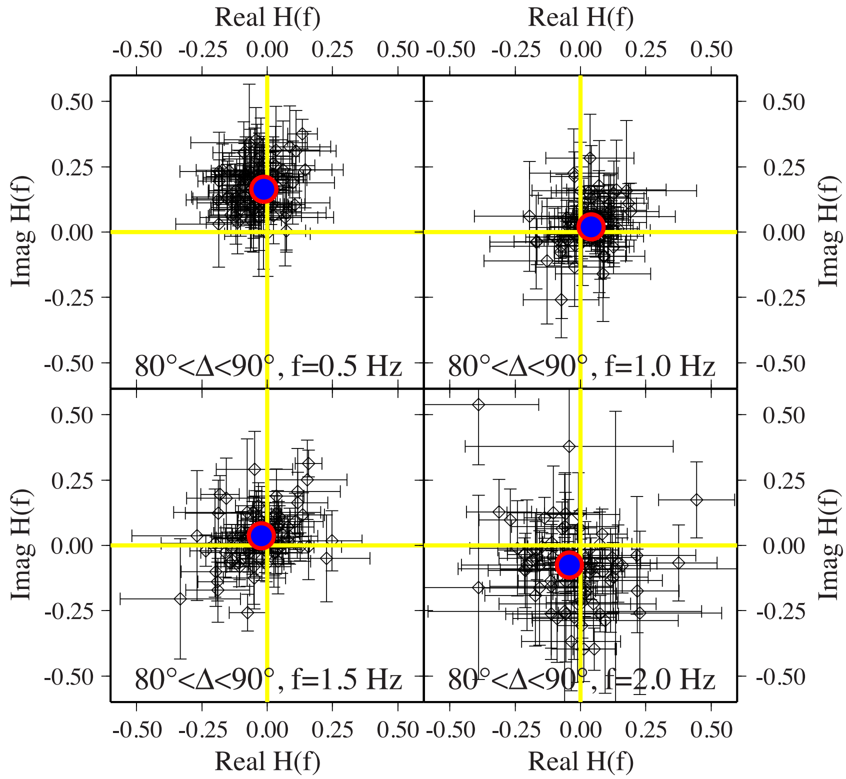

Figure 6. Single-record receiver function estimates have large variance, suggesting the importance of averaging multiple observations. Each panel plots, in the complex plane, single-record estimates of $H_{SV}(f)$ at a single frequency value, from *P* waves recorded at GSN station RAYN. Similar values are expected because the data are from events in a narrow epicentral interval ($80° < \delta < 90°$) and a narrow backazimuth sector (70°–110°). The blue circles indicate the uncertainty-weighted mean estimate of $H_{SV}(f)$. Formal uncertainties in $H_{SV}(f)$ for single-event analysis are indicated by error bars. Formal uncertainties in the weighted sum are 0.006–0.012 for these examples, with error bars much smaller than the plotted symbols.

half-bandwidth pf_R of the multitaper spectrum estimate. We compare the distribution of $\mathcal{S}^2$ values to theoretical χ^2 distributions.

First, we computed the median mod-squared misfit variances $\mathcal{S}^2$ among different frequencies *f* within the various bin-averages. The medians of misfit-variance values for $H(f)$ follow the expectation values of the χ^2-distribution, with some positive bias at larger sample sizes (Fig. 7). These biased cases occur typically where $\overline{H}(f)$ averages data from earthquakes from a broad range of epicentral distance (and incidence angles). Aside from this caveat, we infer that the estimated variances $\mathrm{var}\{H_m(f)\}$ have the proper magnitude.

In contrast to their medians, the histograms of $\mathcal{S}^2$ misfit variances over all frequencies are much broader than theory predicts (Fig. 8). For example, a χ^2-distribution with fewer degrees of freedom (reduced by 30–70 per cent) can represent the width of the sample χ^2 distributions more realistically (Fig. 8). Similar behaviour was found for data sets at several GSN stations.

Two hypotheses for the correlation-of-residuals can be made. First, ambient and signal-generated ground noise at the station may correlate in broad frequency bands among the three seismometer components from event to event, independent of the *P*-wave signal. Because much of this noise is comprised of travelling seismic waves (Shapiro & Campillo 2004), such an underlying noise correlation is plausible. Second, *Ps* converted waves at longer time delays are present within the *P*-coda, as we will show in Section 4. The MTC cross-correlation averages spectral information over a finite frequency interval pf_R, which suppresses the time-domain RF amplitude at $\tau \gtrsim (pf_R)^{-1}$. The leftover 'noise' of the MTC estimate contains this later correlated energy, and may also decrease the effective degrees of freedom in the misfit residuals.

In *LQT* coordinates the coherence between *P* and *SH*-converted energy in the *P* coda is comparable to the coherence between *P* and *SV*-converted energy. For strongly incoherent scattering, it is conceivable that *SV* and *SH* conversions arise from statistically distinct portions of the incoming *P* wave. If *SV* and *SH* scattering are closely linked, simultaneous cross-correlation of all three particle-motion components should lead to RFs that are the same as, or close to, those obtained from pairwise correlations.

We tested the relationship between *P-SV* and *P-SH* scattering by estimating RFs using the SVD. Define a $K \times 3$ matrix $\mathbf{A}(f)$ of eigenspectrum estimates in the *P-SV-SH* coordinate system (in our application $K = 3$):

$$\mathbf{A}(f) = \begin{bmatrix} Y_0^P(f) & Y_0^{SV}(f) & Y_0^{SH}(f) \\ Y_1^P(f) & Y_1^{SV}(f) & Y_1^{SH}(f) \\ & \cdots & \\ Y_{K-1}^P(f) & Y_{K-1}^{SV}(f) & Y_{K-1}^{SH}(f) \end{bmatrix}. \quad (6)$$

For noise-free, perfectly correlated data, the matrix $\mathbf{A}(f) = \mathbf{u}^*(f) \otimes \mathbf{v}(f)$, a single dyad in which the 3-component vector $\mathbf{v}(f)$ contains the correlation coefficients between components. The *K*-component vector $\mathbf{u}(f)$ describes the spectral signature of the correlated signal in a narrow frequency band about *f*. The RFs $H_{SV}(f)$ and $H_{SH}(f)$ correspond to the component ratios v^{SV}/v^P and v^{SH}/v^P, where $\mathbf{v}(f) = [v^P, v^{SV}, v^{SH}]$.

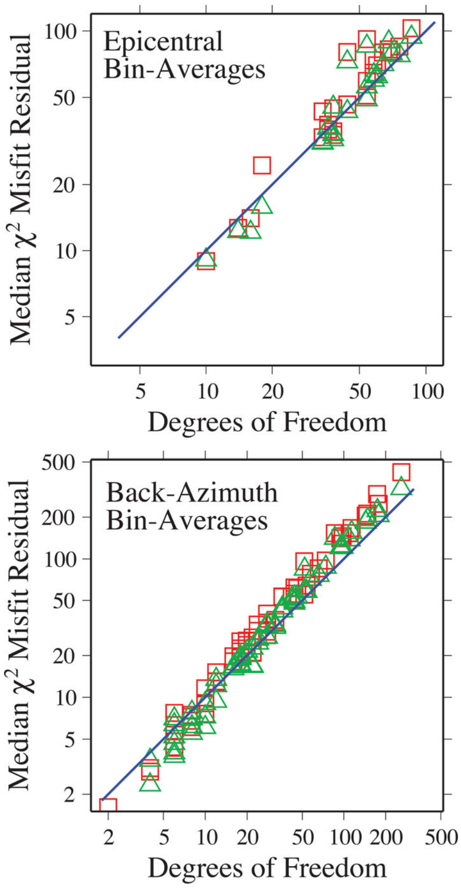

Figure 7. Median misfit-variance residuals, in the frequency domain, for the bin-averaged MTC receiver functions for GSN station RAYN displayed in Figs 2 and 3. Squares and triangles plot medians of misfit variance from the *P-SV* and *P-SH* receiver function estimates, respectively. A bin average of MTC RFs that uses *P*-waves from *M* events has misfit-variance residuals whose expected $\chi^2 = 2M - 2$. The sample distribution of misfit-variance residuals is drawn from bin averages computed at evenly spaced frequencies of the discrete Fourier transform (DFT) for $f < 4.0$ Hz. Similar results are obtained if the computation is restricted to $f < 2.0$ Hz. This figure suggests that the formal uncertainties of the MTC RFs do not greatly over- or underestimate the *a posteriori* sample uncertainties for $H(f)$.

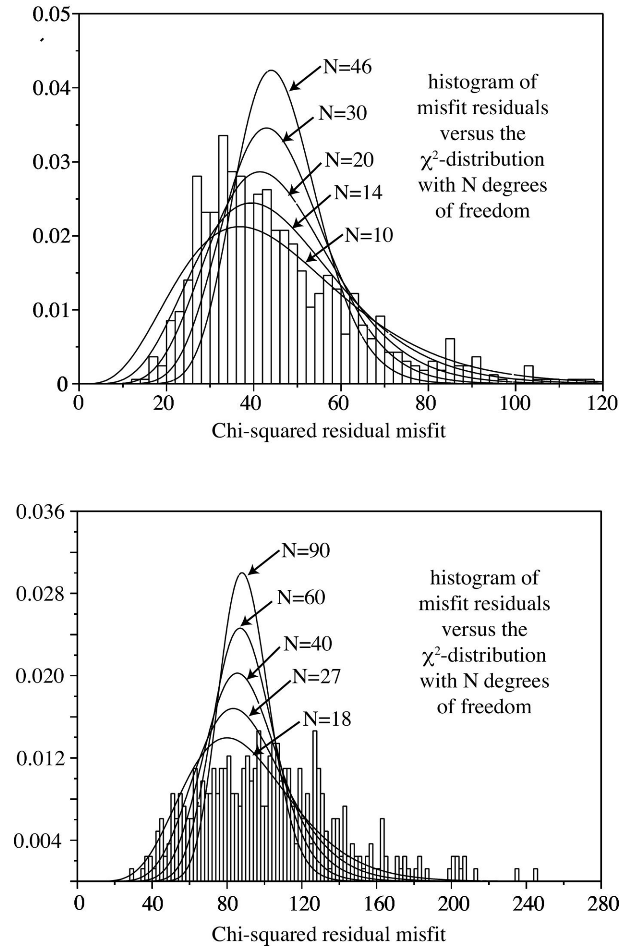

Figure 8. The misfit-variance residuals, in the frequency domain, for two bin-averaged MTC receiver functions for GSN station RAYN displayed in Figs 2 and 3. Bin stacks with $M = 24$ and $M = 46$ events are compared, whose residuals would follow the χ^2-distribution with $N = 2M - 2$ degrees of freedom if the demeaned observations were Gaussian and uncorrelated, that is, $N = 46$ and $N = 90$ respectively. The graphs indicate that a much smaller value of N describes the sample distribution of χ^2 residuals better than the nominal value of N.

We associate the SVD RF with the singular vectors of **A** that correspond to its largest singular value, similar to the 'pure state' filtering algorithm (Samson 1983). We decompose

$$\mathbf{A}(f) = \sum_{j=1}^{3} \lambda_j(f)\mathbf{u}_j^*(f) \otimes \mathbf{v}_j(f), \tag{7}$$

where $\lambda_1 > \lambda_2 > \lambda_3$ are the singular values. We estimate the uncertainty of the SVD RF by analogy to eqs (3) and (4). We identify 'coherence' as the fractional variance explained by the pure-state singular vector,

$$C_{P\text{-}SV}(f) = \frac{\left(\lambda_1^2 v_1^{SV}(f)\right)^* v_1^P(f)}{\left(\sum_{k=0}^{K-1}\left(Y_P^{(k)}(f)\right)^* Y_P^{(k)}(f)\right)^{\frac{1}{2}}\left(\sum_{k=0}^{K-1}\left(Y_{SV}^{(k)}(f)\right)^* Y_{SV}^{(k)}(f)\right)^{\frac{1}{2}}}. \tag{8}$$

We can use eq. (3) to estimate RF variance with only a change of subscripts:

$$\text{variance}\,(H_{SV}(f)) = \left(\frac{1 - C_{P\text{-}SV}^2(f)}{(K-1)C_{P\text{-}SV}^2(f)}\right)|H_{SV}(f)|^2. \tag{9}$$

Similar formulas define the SVD RF for the *SH* component.

In tests with RAYN data, the added sophistication of the SVD RF algorithm leads only to small changes in the estimated RFs. We infer that *P-SV* conversion and *P-SH* conversion in the *P* coda are co-dependent, in a statistical sense. It should therefore be feasible to estimate $H_{SV}(f)$ and $H_{SH}(f)$ jointly. Although our experiments suggest that the ordinary pairwise MTC RF estimator is adequate in practice for *Ps* converted phases, the SVD-based RF estimator may have utility for reconstructing *Sp* converted phases. The estimation of *S* RFs can encompass more than one impinging polarization, in principle.

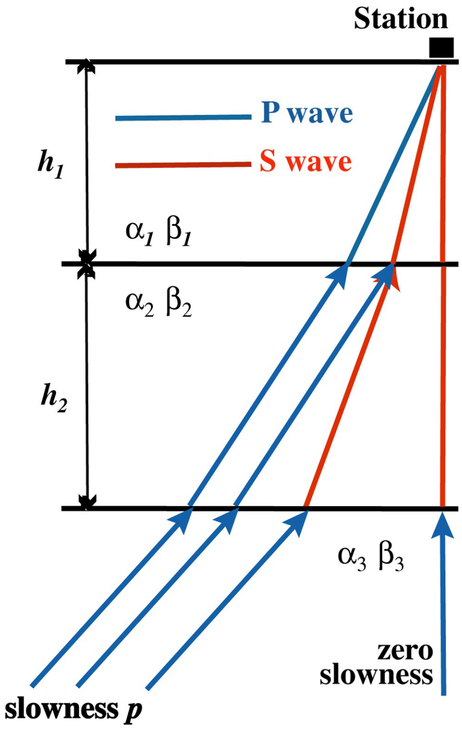

Figure 9. Schematic diagram of ray paths used in the migration of receiver function estimates.

4 RECEIVER-FUNCTION MOVEOUT IN THE FREQUENCY DOMAIN

Although RFs are ideally computed from *P*-waves at near-vertical incidence, in practice we often combine data from earthquakes at widely disparate epicentral distances, across a range of incidence angles. The time delay of a *Ps* converted phase varies with incidence angle, and this variation increases as the converting interface deepens. In order for *Ps* converted phases from, say, a steeply incident *PKP* phase and a shallow-incident near-teleseismic *P* to add coherently, we must correct the RFs for timing differences imposed by incidence-angle variation. Published RF-migration algorithms (Sheehan *et al.* 2000; Morozov & Dueker 2003a,b; Poppeliers & Pavlis 2003a,b) typically apply a timing correction after the time-domain RF $H(t)$ is estimated from a single *P*-wave coda. Because the uncertainty of the MTC RF is estimated in the frequency domain, it is advantageous to correct moveout and sum $H_{SV}(f)$ and $H_{SH}(f)$ before converting to the time domain.

The following algorithm was first described by Leahy & Collins (2009). Assume a 1-D seismic wave-speed model for the upper layers of Earth beneath a station. Used only for stacking, this model can be composed of *M* constant layers over a half-space (Fig. 9), much simpler than the likely velocity structure beneath the station. Refraction causes teleseismic seismic body waves to steepen as wave speed decreases in Earth's crust. As a result, *Ps* converted waves from crustal interfaces require a smaller moveout correction than do conversions from mantle interfaces. Consider first the top layer of the stacking model, with *P* and *S* wave speeds α_1 and β_1. For a vertically incident wave, the *Ps* delay time for an interface at depth *h* within the top layer is

$$t_{PS} = \frac{h(\alpha_1 - \beta_1)}{(\alpha_1\beta_1)}. \tag{10}$$

Relative to vertical incidence, the delay time of a *Ps* converted wave with slowness *p* in the top layer of the stacking model is stretched by a factor of

$$\gamma_1 = \frac{\alpha_1\cos\phi_1 - \beta_1\cos\theta_1}{\alpha_1 - \beta_1}, \tag{11}$$

where θ_1 and ϕ_1 are the angles of incidence for the *P* and *S* waves, respectively, $\cos\theta_1 = \sqrt{1-\alpha_1^2p^2}$ and $\cos\phi_1 = \sqrt{1-\beta_1^2p^2}$. The time-scaling factor γ_1 for *Ps* delay can be represented in the frequency domain by a reciprocal scaling factor γ_1^{-1} for frequency. More precisely, the time scaling $H(t_{PS}) \rightarrow \tilde{H}(\tilde{t}_{PS}) = H(\gamma_1 t_{PS})$ corresponds to a frequency scaling $H(f) \rightarrow \tilde{H}(\tilde{f}) = H(f/\gamma_1)$. We effect this scaling in practice by numerically interpolating $H(f)$ and its uncertainty estimate on a sequence of discrete frequencies $\{\tilde{f}_j\}$ that lie between the frequencies of the DFT.

We 'stretch' the frequency-domain RF by interpolating its frequency points at a spacing Δf smaller than that of the DFT. This causes the time-domain RF to 'shrink' to simulate a *Ps* conversion at the delay time appropriate for vertically travelling waves. If we 'compressed' the frequency-domain RF by interpolating its frequency points at a spacing Δf larger than that of the DFT, the time-domain RF would stretch so that *Ps* pulses arrive at longer delay times. We stack RFs from independent *P* waves using the interpolated $H(f)$, preserving the relative weighting afforded by the interpolated MTC uncertainty estimates. We compute a time-domain RF via inverse DFT that places *Ps* converted phases at time delays appropriate for a vertically incident *P* wave.

Because the scaling factors γ_j vary among layers, there is no single reciprocal scaling in the frequency domain that adjusts *Ps* conversions throughout the stacking model. Instead, we correct the moveout in a piecewise manner, once for each layer in the stacking model, using the sequence of scaling factors $\gamma_j, j = 1, \ldots M+1$. (The index $j = M + 1$ denotes the half-space at the base of the stacking model.) For each seismic record, we perform $M + 1$ spline interpolations of the estimated RF. Each bin-averaged $H(f)$ is estimated $M + 1$ times, and subjected to an inverse DFT to create a time-domain RF estimate. Spline interpolation and FFTs are not computationally burdensome, so this migration scheme is quite fast. The final step is a splice of the $M + 1$ time domain RFs to form a single time-domain RF, defined by tie points that correspond to the boundaries of the stacking model. In particular, if stacking-model layer thicknesses are $h_1, h_2, \ldots h_M$, the time delay for a vertically incident wave that corresponds to the transition depth between Layers 1 and 2 is $\tau_1 = h_1(\alpha_1 - \beta_1)/(\alpha_1\beta_1)$. The time delay between Layers *j* and $j - 1$ is $\tau_j = \tau_{j-1} + h_j(\alpha_j - \beta_j)/(\alpha_j\beta_j)$. The time delay for a *P* wave of slowness *p* that corresponds to the transition depth between Layers 1 and 2 is $\tilde{\tau}_1 = \gamma_1\tau_1$, and that between Layers *j* and $j-1$ is $\tilde{\tau}_j = \tilde{\tau}_{j-1} + \gamma_j(\tau_j - \tau_{j-1})$. Alignment of the $M + 1$ independently corrected RF at the designated tie points is effected by multiplication of the frequency-domain RFs by complex phase factors $\exp(-i2\pi f\xi_j)$ before the inverse DFT. The phase correction for the *j*th time-segment is

$$\xi_j = \begin{cases} 0 & \text{for } j = 1 \\ \tilde{\tau}_{j-1}/\gamma_j - \tau_{j-1} & \text{for } j > 1 \end{cases}. \tag{12}$$

Moveout correction is an important component of RF migration, which improves the alignment of major *Ps* converted phases in MTC bin-stacks, most noticeably adjusting the alignment of strong Moho-converted phases from shallow-incident *P* waves (Fig. 10). The discrepancy between RFs migrated with the upper-crust seismic wave speeds and uppermost-mantle wave speeds is modest but

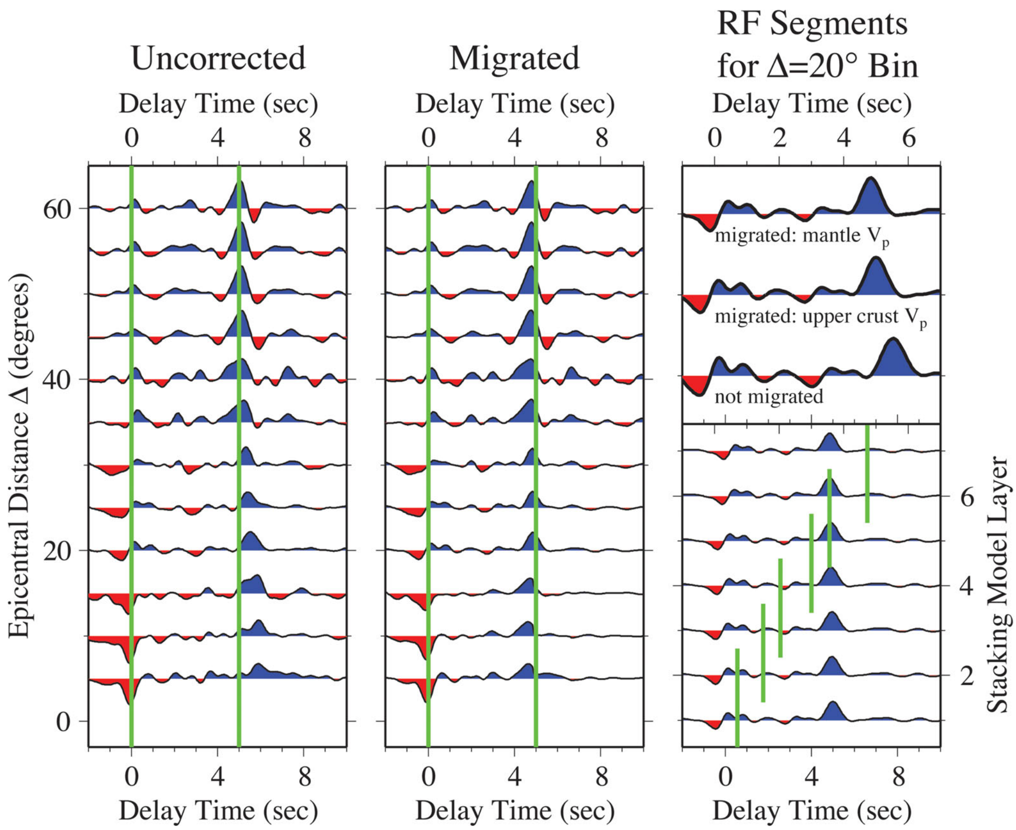

Figure 10. Migration of *P-SV* receiver functions in the 50°–110° backazimuth sector for GSN station RAYN, using events with magnitude $M > 6.0$, a 60-s time window, and 5° half-width in Δ. The stacking model is an isotropic version of the seismic velocity model proposed by Levin & Park (2000) for the crust and uppermost mantle beneath this station. The migrated delay time t is appropriate for a P wave arriving at vertical incidence. The left panel is not corrected for moveout, causing misalignment of the Moho-converted *Ps* phase, for which the vertical green line references the expected delay time. The centre panel is corrected for moveout, and aligns the *Ps* phases better for near-teleseismic events. The lower right panel shows the effects of interpolating the spectra of P coda to compensate for moveout in different layers of the stacking model, using earthquakes at epicentral distances $15° < \Delta < 25°$. Each trace is valid for a delay-time interval that lies between splice points indicated by vertical green lines. The upper-right panel compares the migrated RF traces for the upper crust and uppermost mantle to the unmigrated RF for earthquakes at epicentral distances $15° < \Delta < 25°$.

significant (Fig. 10, right panel). A larger discrepancy is seen when compared with the unmigrated RF. Moveout-correction potentially improves the retrieval of higher-frequency signals in time-domain RFs, as small timing misalignments can disrupt the signal-to-noise enhancement associated with stacking. In practice, migration of *Ps* from crustal interfaces induces only modest improvements in bin-averaged RFs unless near-teleseismic ($\Delta < 30°$) earthquakes comprise a substantial fraction of the data set.

There is no mathematical guarantee that differently migrated frequency-domain RF bin-gathers will produce time-domain RFs that match at their predicted splice points. In most examples with real data, the corrected time-domain RFs appear to vary smoothly across the splice points τ_j for cut-off frequencies $f_c \lesssim 1$ Hz. For higher frequencies, however, discontinuities in the derivative of the RF can be noticeable at tie points associated with sharp velocity jumps, for example, the Moho. Sometimes a tie point suffers a first order discontinuity. A smoother transition can be effected by averaging the migrated RFs in a short interval across the splice.

5 EVENT STACKING FOR *Ps* CONVERSIONS AT LONG DELAY TIMES

The major shortcoming of the MTC RF estimator of Park & Levin (2000) is its difficulty retrieving *Ps* conversions at long delay times. RF studies of upper-mantle structure at the base of the lithosphere and the transition zone involve *Ps* converted phases with delay times between 20 and 70 s (e.g. Bostock 1998; Sheehan *et al.* 2000), but the frequency-averaging properties of the Slepian data tapers with time-bandwidth product $p \geq 2$ tends to limit typical RFs to delay times $\tau < 20$ s. Frequency-averaging affords the MTC algorithm desirable statistical properties, especially for single-event RF estimates. We can achieve a narrower frequency average for the retrieval of *Ps* from interfaces deeper than 100 km if we sum single spectral estimates over K seismic events, rather than K eigenspectra from one event, in the RF cross-correlation formulae (2) and (3). Using only one spectrum estimate per component for each P wave, we estimate the spectra with a Slepian taper of time-bandwidth $p = 1$, that is, the 'one-π-prolate' taper. This taper averages local frequency information over a central bandwidth similar to the simple boxcar taper, but is optimized for spectral leakage resistance.

For GSN station RAYN, the use of 'event-based' cross-correlation (ECC) to estimate $H_{SV}(f)$ and $H_{SH}(f)$ extends the range of the time-domain RF to allow easy recognition of the first crustal reverberations $Pp_{MS}s$ and $Ps_{MS}s$ at ~20-s delay, and signals that are plausibly associated with *Ps* conversions from the 410- and 670-km discontinuities (Figs 11 and 12). Compared with the standard MTC RF estimator, the ECC RF estimator suffers a greater likelihood that the number and quality of earthquakes in an interval of

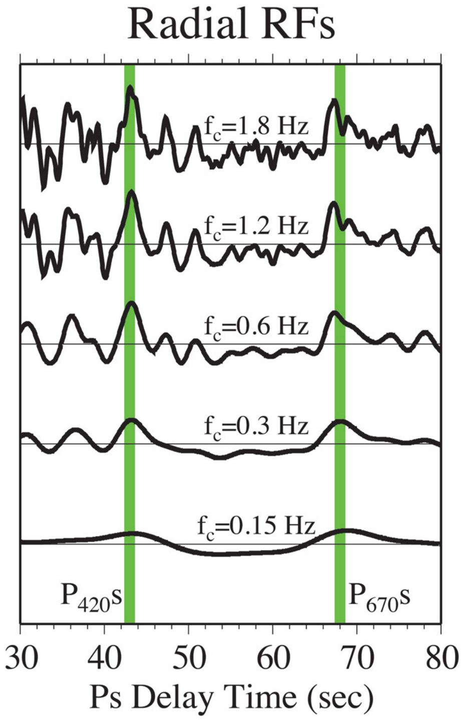

Figure 11. Radial-component *Ps* converted phases associated with the mantle transition zone beneath GSN station RAYN, computed from 160-s time windows with event stacking for earthquakes at $70^\circ \leq \Delta \leq 90^\circ$ in the 50°–110° backazimuth sector. Lowpassed radial-horizontal components are shown, with corner frequencies labelled. For the lowest bandpass (0.15 Hz) the peak-to-peak *Ps* amplitude from the transition-zone discontinuities is 0.3–0.4 per cent of the direct-*P* amplitude. For the highest bandpass (1.8 Hz) the peak *Ps* amplitudes are 0.8 per cent and 0.6 per cent of direct *P* for the 420 and 670 km interfaces, respectively.

backazimuth or epicentral distance are insufficient to obtain a stable RF estimate. Another shortcoming arises from the seismic wavefield itself. For many seismological stations, core-refracted *PKP* phases are important sources of data coverage, for instance, at GSN station RAYN nearly all large events in the backazimuth range 120°–180° contribute *PKP* waves rather than *P* waves that bottom in the mantle. For a broad range of epicentral distance, unfortunately, the surface-reflected *PP* phase follows the core-refracted *PKP* phase closely in time, obscuring potential late-arriving *Ps* conversions within the *PKP* coda.

Moveout-correction is potentially an important factor in the retrieval of late-arriving *Ps* via ECC. The frequency-domain stretching outlined in the previous section can be applied to ECC RF estimates. A stacking model for the upper mantle defines scaling factors for time-delay perturbations and the interpolation interval for spectral estimates $Y_P^{(k)}(f)$, $Y_{SV}^{(k)}(f)$ and $Y_{SH}^{(k)}(f)$, based on the *P*-wave slowness associated with the *k*th seismic record in a cross correlation. We improve considerably the resolvability of transition-zone converted phases $P_{410}s$ and $P_{670}s$ (Fig. 13). (Note that the retrieval of $P_{410}s$ and $P_{670}s$ is not perfect, even after correction. The deeper *Ps* phase is less prominent and varies with backazimuth.)

6 MOVING WINDOW STACKING FOR *Ps* CONVERSIONS AT LONG DELAY TIMES

An alternate moveout-correction scheme for the MTC RF-estimator involves time-shifts of data windows prior to multiple-taper cross-correlation (Helffrich 2006; Shibutani *et al.* 2008). With a typical time-bandwidth $p = 2.5$ and $K = 3$ Slepian tapers, if 60-s time windows of the *P* and *SV* components of motion are correlated, the MTC RF reconstructs potential *Ps* converted phases at time delays $-12\ \text{s} \lesssim \tau \lesssim 12\ \text{s}$ (pulses at negative time delays would be acausal). If the 60-s *P*-component record is cross-correlated with a 60-s *SV* record that starts 10-s later, an implicit phase shift

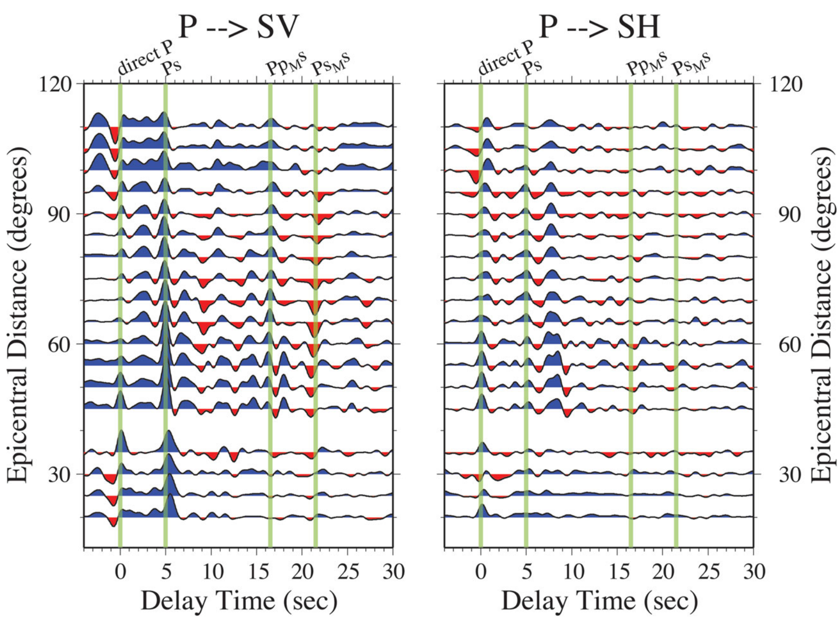

Figure 12. Epicentral sweeps of event-stacked RFs for GSN station RAYN using a single 1π-prolate taper to analyse 160-s time windows of *P*-coda from earthquakes at epicentral distances $\Delta \leq 120^\circ$ in the 50°–120° backazimuth sector. We filter the RFs below $f_C = 1.0$ Hz to avoid bandedge ringing via a squared-cosine taper in the frequency domain, with half-amplitude at 0.5 Hz. The vertical lines indicate the expected delay times for Moho-converted *Ps*, and reverberations Pp_Ms and Ps_Ms. Pp_Ms and Ps_Ms are predicted to have positive and negative polarity, respectively, in agreement with the estimated RFs.

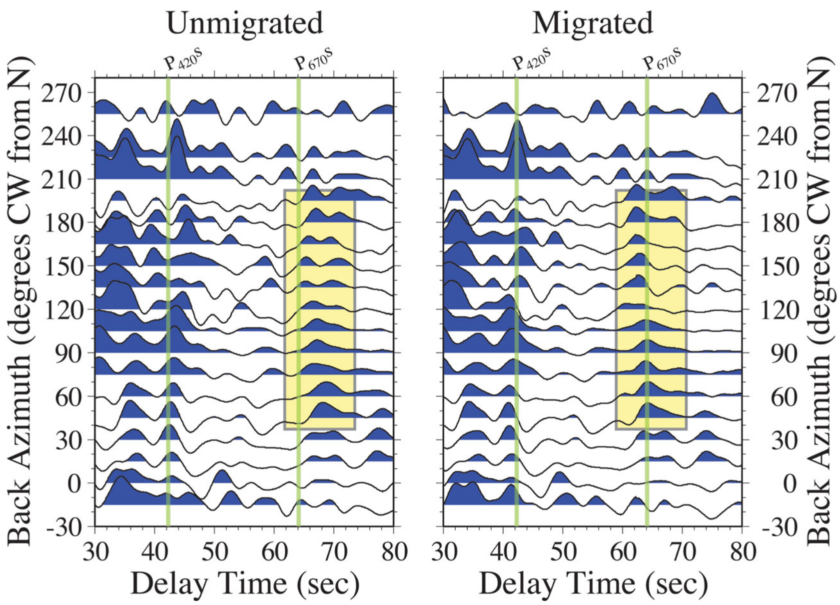

Figure 13. Frequency-domain migration applied to backazimuth sweeps of event-stacked RFs for GSN station RAYN using a single 1π-prolate taper to analyse 160-s time windows of P-coda from earthquakes at epicentral distances $40° \leq \Delta \leq 120°$. We bandpass the RFs below $f_C = 0.4$ Hz, and avoid bandedge ringing via a squared-cosine taper in the frequency domain, with half-amplitude at 0.25 Hz. The vertical lines mark the expected time delays of $P_{410}s$ and $P_{670}s$ for vertical incidence. The yellow boxes show how frequency-domain moveout corrections align 'multiple' expressions of the $P_{670}s$ converted wave into a single pulse.

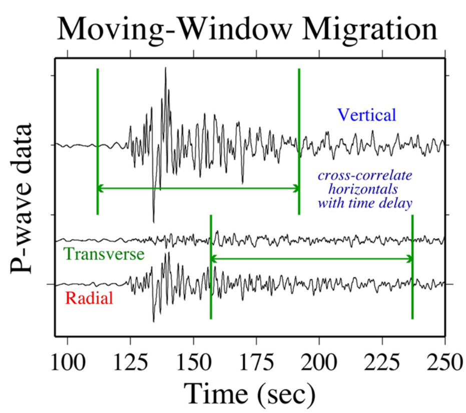

Figure 14. Illustration of data windows used in a moving-window migration (MWM) RF estimate centred on the expected arrival of the Ps converted phase from the 410-km discontinuity. Note that the vertical-component data window starts roughly 10 s before the P-wave onset. We advise incorporating an interval of pre-event noise in practice, rather than restricting the data window to only the P coda, because the Slepian data tapers downweight the extremes of the data interval to resist spectral leakage. See the Supporting Information for a more detailed discussion of spectrum-analysis formalities.

in the frequency domain causes the MTC RF to reconstruct Ps converted phases at -2 s $\lesssim \tau \lesssim 22$ s. The time windows of the P and SV components need not overlap in time, as long as the cross-correlation involves components produced by the same earthquake (Fig. 14). Moveout for a particular Ps conversion can be corrected by time-shifting the SV and SH time windows, relative to the P time window, in accordance with the time delay predicted for the phase.

We call this algorithm moving-window migration (MWM). Helffrich (2006) calls the algorithm 'Extended Time Multiple Taper Receiver Function', or ET MTRF. Whatever the acronym, the algorithm offers the ability to resolve Ps from mid-mantle interfaces with the statistical benefits of the multiple-taper approach. Fig. 15 shows a sweep of MWM RFs, binned in intervals of epicentral distance, focussed on the predicted moveout for a Ps conversion from a discontinuity at 220-km depth. The largest signal in the figure is a negative-polarity Ps_Ms crustal reverberation, distinguished by 'negative' moveout, that is, smaller time delays at smaller epicentral distances. The MWM pre-DFT time shifts follow the 'positive' moveout of a direct $P_{220}s$ conversion, assuming such a converted phase is in the data. Signals from seismic phases that share the predicted moveout will be enhanced by stacking, relative to noise, and appear as a correlated pulse with no moveout in the epicentral RF gather.

A candidate for a Ps phase is apparent in Fig. 15, exhibiting consistent amplitude for $60° \leq \Delta \leq 90°$. The small time delay relative to 'zero' indicates that the potential converting interface would lie somewhat deeper than the target depth, near 235 km. Moving-window migration can be used to track Ps reverberations as well as direct Ps conversions (Fig. 15). For this data set, the reverberations are not evident for near-teleseismic events ($\Delta < 30°$).

The algorithm described by Helffrich (2006) computes evenly spaced segments of the time-domain RF for a single event recorded at a single station. The Helffrich (2006) algorithm splices a composite RF for each event and station, and afterwards stacks multiple records in the time domain, correcting for moveout by interpolating the time variable of the composite RF. Our algorithm stacks MWM RFs from multiple events in the frequency-domain because this allows us to apply inverse-variance weighting of $H_{SV}(f)$ and $H_{SH}(f)$ estimates in the stack. Our implementation of moving-window

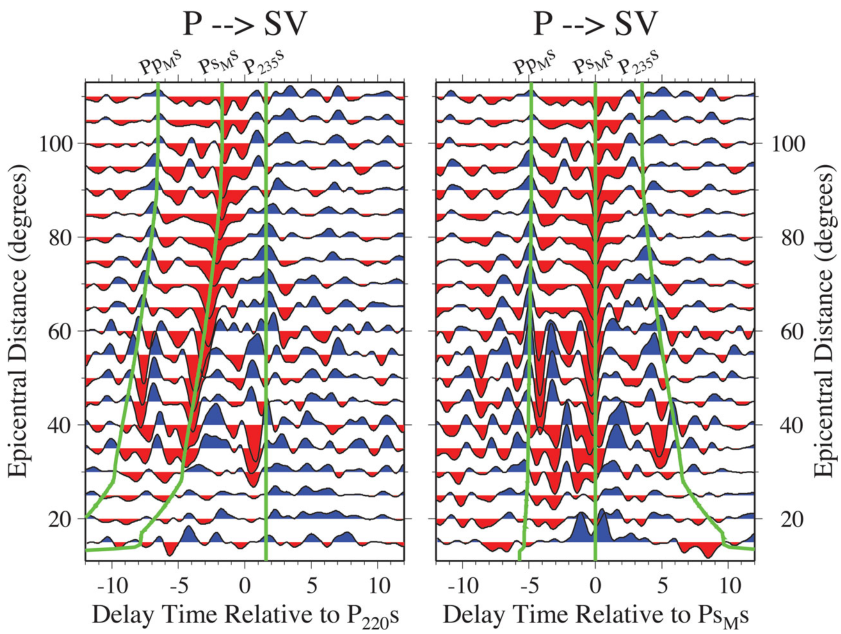

Figure 15. Left panel: epicentral bin averages (half-width 5°) of moving-window migration (MWM) RF estimates targeted at 220-km depth in Earth model IASPEI91 (Crotwell *et al.* 1999). The Q-component RFs are shown to reconstruct *P*-to-*SV* converted waves. We use three 2.5π-prolate Slepian tapers to analyse 60-s time windows of *P*-coda from earthquakes in the 50°–110° backazimuth sector, recorded at GSN station RAYN. We filter the RFs to avoid bandedge ringing via a squared-cosine taper in the frequency domain, with $f_c = 1.4$ Hz. The vertical line indicates the expected timing of *Ps* from an interface at 235-km depth. The curved lines indicate the expected arrivals of crustal reverberations Pp_{MS} and Ps_{MS} for the IASPEI91 model with a thicker 38 km crust and slower lower-crust $\beta = 3.6$ km s^{-1}. Right panel: Epicentral bin averages (half-width 5°) of moving-window migration (MWM) RF estimates targeted at the Ps_{MS} reverberation phase in our modified IASPEI91 model. Other analysis parameters are equal to those used for the left-panel RFs.

migration defines RF segments by a series of target depths, and relates them to delay times (computed for a reference model) for a vertically incident *P* wave. For each RF segment at a specified target depth, we compute moveout corrections for each seismic event recorded at a particular station, and compute MTC RFs with *P* data series and the time-shifted *SV* and *SH* data series. We stack the frequency-domain RFs with inverse-variance weighting, and compute overlapping time-domain RF segments via the inverse DFT. These RF segments have already been corrected for moveout to their target depths. We splice the segments into a composite extended-time RF by applying an overlap taper to smooth the RF-segment splice points.

Because FFTs are computationally quick, the need to recompute RFs for a variety of depth ranges is not a great burden. Fig. 16 shows examples of such composite MWM RFs, migrated for *Ps* phases for evenly spaced target depths every 40 km between 0- and 680-km. Bin-stacked RFs for successive target depths are spliced with some overlap, using a weighted average to smooth the transition. The agreement between event-stacking with frequency-domain moveout correction (Fig. 13) and MWM is encouraging. Because the MWM algorithm can be run separately on each of the many *Ps* conversions, it is potentially more flexible, allowing the analyst to test hypotheses about an isolated *Ps* conversion with less computation. Frequency-domain RF moveout-correction uses a 1-D seismic-velocity model with a small number of constant layers. The MWM algorithm can use raytracing in a continuously varying wave speed model to predict data-window time delays. The raytracing option may be helpful for deeper interfaces with *Ps* conversion points that are widely separated, because 2-D and 3-D structure may exert greater influence on *Ps* moveout.

7 JACKKNIFE UNCERTAINTY FOR RECEIVER FUNCTIONS IN THE TIME DOMAIN

Formal uncertainties for MTC RFs are expressed in the frequency domain, where the standard deviation of $H(f)$ estimates varies with the coherence of the *P*, *SV* and *SH* motion components. When translated into the time domain, however, the formal uncertainty of the MTC RF is constant with time, as a consequence of assuming that the sample residuals of $H(f)$ about its mean are uncorrelated. *A posteriori* statistical tests of the $H(f)$ residuals indicate that correlation is probably significant. We therefore seek an alternative to formal uncertainty estimates.

A more flexible indicator of uncertainty in the time-domain RF can be obtained with nonparametric resampling methods such as the jackknife (Efron 1982). The uncertainty of $H(t)$ can be estimated from the scatter of RF-estimates made from subsets of the available data. In the jackknife, one uses *M* subsets of an *M*-point data set. Each data subset has one of the data points deleted. For RF estimation over an interval of epicentral distance or backazimuth, we jackknife over events, so that an bin-average with 10 events leads to 10 different RF estimates, an interval with 45 events leads to 45 different RF estimates, and so on. We apply formulas from Efron (1982) for computing the jackknife standard deviation of $H(t)$ from the time-variable scatter of data-subset estimates for $H(t)$.

We apply the jackknife to the epicentral bin-averages of MWM RFs computed for GSN station RAYN (Fig. 17), adjusted for *Ps* conversions at the depth identified with the Lehmann discontinuity (Anderson 2011). Two features of comparison with the simple RF estimate are instructive. First, the isolated bumps and wiggles at migrated $\tau \approx 0$ s for $\Delta < 60°$ are revealed by the jackknife to be

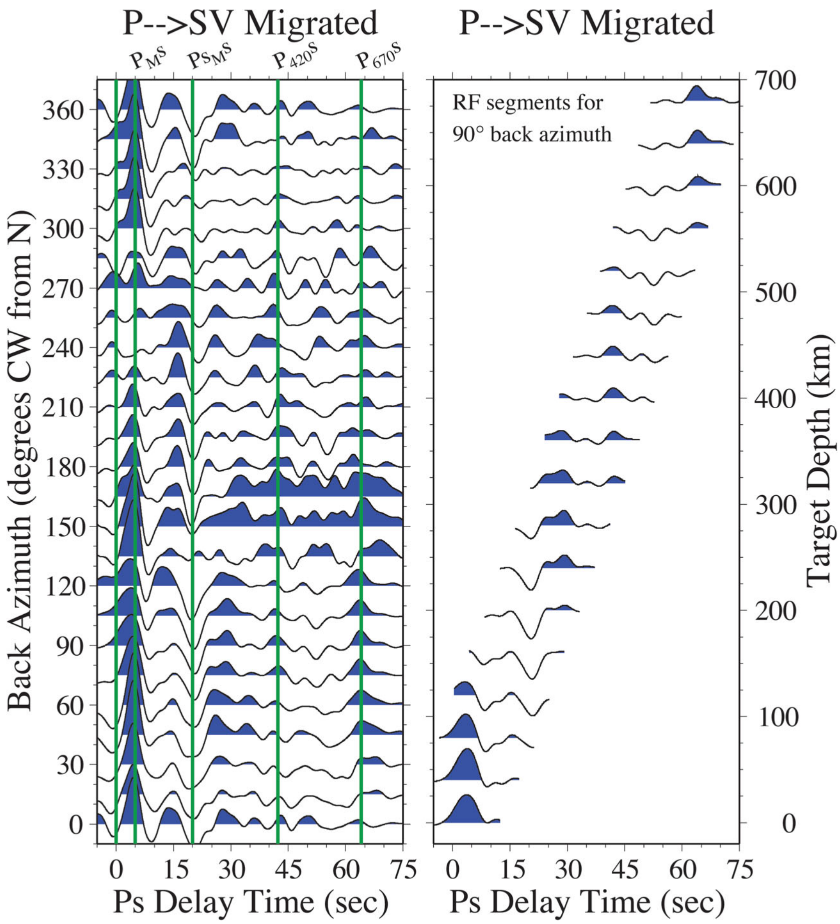

Figure 16. Left panel: backazimuth bin averages (half-width 15°) of moving-window migration (MWM) RF estimates spliced from a succession of RF estimates at target depths 0, 40, 80, 120, . . . 640 and 680 km. Delay time is referenced to a vertically incident P wave. We use three 2.5π-prolate Slepian tapers to analyse 60-s time windows of P-coda from 720 earthquakes recorded at GSN station RAYN, with RF estimation limited to $f_c \leq 0.3$ Hz. Note that the reverberation Ps_{MS} is strongly evident as a negative pulse at $t \approx 20$ s, but its shape is distorted because the MWM moveout does not select for it. Note that $P_{410}s$ is stronger for easterly backazimuths (180°–360°), compared with westerly P arrivals (0°–180°). Although P waves are less plentiful from the west, the stronger recovery of $P_{670}s$ for westerly backazimuths suggests that the backazimuth dependence of $P_{410}s$ is not caused by incomplete migration. Right panel: for earthquakes in the backazimuth sector 75°–105°, overlapping RF segments for target depths 0, . . . 680 km. Note that the largest amplitude of a major RF feature, such as $P_{410}s$, occurs for the RF segment that targets a nearby depth.

poorly constrained by data. Second, the pulses that could represent a Ps conversion near 220-km depth, appear to be robust to data resampling, even though observed over a restricted range of Δ. Note that the crustal reverberations Pp_{MS} and Ps_{MS} also are robust over a limited epicentral interval, losing amplitude at small Δ. A small jackknife uncertainty does not prove the existence of a Lehmann-converted Ps phase beneath RAYN, but such an identification is nonetheless bolstered.

8 SUMMARY AND BROADER IMPACTS

This report presents several extensions and tests of algorithms for estimating teleseismic RFs $H(f)$ with multiple-taper spectral analysis (Table 1). Computer codes to implement the algorithms described here can be accessed at http://jparkcodes.blogspot.com. Originally developed to deconvolve the radial and transverse components of P-wave motion with the vertical component (Park & Levin 2000), the MTC RF estimate translates readily to a ray-based LQT coordinate system to estimate P-SV and P-SH conversion more directly. Using data sets from several permanent seismic observatories of the GSN network, we find that the average P-SV and P-SH coherences do not differ greatly, suggesting that both types of body-wave scattering make a systematic contribution to the P coda. Extension of pairwise cross-correlation to multicomponent correlation based on the SVD indicates that P-SV and P-SH scattering are strongly correlated, because both can be retrieved from one frequency-dependent 'pure state' singular vector.

The MTC RF estimate has a frequency-dependent formal uncertainty that trends inversely with the coherence between components. *A posteriori* tests of these uncertainty estimates, based on the misfit residuals of RF estimates $H(f)$ averaged over narrow intervals of epicentral distance and backazimuth, show that their median χ^2 values agree with the expected value of the χ^2-distribution in most cases. However, the histograms of χ^2 residuals for RF estimates

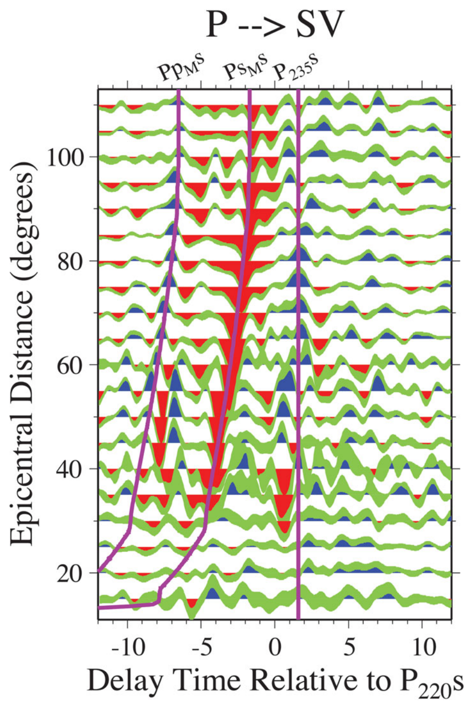

Figure 17. Jackknife uncertainties for bin-averaged MWM RF estimates, using an epicentral sweep centred on the 220-km-depth discontinuity identical to Fig. 15. The uncertainty of the time-domain RF is indicated by the green interval about the bin-averaged RFs. Blue and red area fills indicate positive and negative RF excursions that are robust to jackknife uncertainties.

at different frequency points are much broader than the nominal degrees of freedom would indicate. Broader misfit-variance histograms suggest that the uncertainties of MTC RF estimates from different earthquakes are correlated, not statistically independent. We interpret this correlation as arising from *Ps* converted phases at delay times outside the narrow time interval that is reconstructed by the MTC RF.

To improve resolution of late-arriving *Ps* converted energy, we describe (1) a frequency-domain algorithm to compensate for *Ps* moveout, (2) an alternate event-based stacking of single-taper spectrum estimates that extends the range of delay times that can be resolved using the frequency-domain approach, and (3) a moving-window migration algorithm that reconstructs *Ps* converted energy at long delay times while retaining the statistical behaviour of the original MTC RF estimator. Each approach has its merits, but the moving-window migration may be most flexible, because *Ps* moveout could readily be computed from 2-D and 3-D wave speed models with a small alteration in the code. Finally, we have developed a jackknife uncertainty estimator for $H(t)$ that offers a time-dependent estimate of uncertainty in the RF.

Finally, the MTC RF estimator can be adapted beyond single-station *Ps* converted-wave reconstruction and 1-D structure. Starting with a collection of seismic events distributed widely in backazimuth, one can stack the frequency-domain RFs $H(f)$ with harmonic weighting to infer the influence of anisotropy and dipping interfaces (Shiomi & Park 2008; Bianchi *et al.* 2010; Liu *et al.* 2015; Olugboji & Park 2016; Park & Levin 2016). Bianchi *et al.* (2010) and Liu *et al.* (2016) have used the correlation and coherence estimates of MTC RFs to stack data with uncertainty weighting from large-N arrays of stations and create lateral transects of composite RFs to infer 3-D interface structure beneath Italy and NE Asia, respectively. It is possible to extend MTC to other scenarios of body-wave scattering, for example, *Sp* converted phases and *SS* precursors. The ambiguity of initial *S*-wave polarization (*SH* or *SV*) will likely motivate an adaptation of the SVD-based RF estimator discussed in Section 3. MTC, and its associated uncertainty estimates, may also inform useful updates to other Fourier-domain methods. For example, Prieto *et al.* (2007) apply multiple-taper methods, including the jackknife, to constrain earthquake parameters that depend on ground-motion spectra. Further applications include inferring surface-wave dispersion from ambient noise (Ekstrom *et al.* 2009), distinguishing waveforms from similar or repeated earthquakes with a transfer-function metric (Kennett & Fichtner 2012), and detecting small seismic events in noise (Gibbons *et al.* 2008).

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Table 1. Multiple-taper correlation (MTC) algorithms for receiver-function (RF) estimation. The techniques are not exclusive, but can be combined in some cases. For instance, one could implement moving-window moveout corrections in an SVD-based RF estimate.

Multiple-taper RFs with no moveout correction	Frequency-domain moveout correction	Moving-window moveout correction	SVD-based correlation RF estimates	Event-based correlation RF estimates
ZRT or *LQT* correlations, one $H(f)$ estimate per event	Interpolate $H(f)$ estimates with variable Δf for each model layer	Time-shift *R*, *Q*, or *T* data window to predicted delay time of *Ps* phase	Estimate $H(f)$ from multi-component correlation, rather than pairwise	Estimate *ZRT* or *LQT* correlations from event gather or array gather
Good for short delay times, crustal structure, first-look analysis	Focus on late-arriving *Ps*, mantle structure, tuning models	Late-arriving *Ps*, good for phases with non-standard moveout	Good for phases with ambiguous polarization, for example, *Sp* converted waves	Good for long data intervals, large-N array correlation

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this paper:

Figure S1. Left panels: The first $K = 3$ Slepian tapers for time-bandwidth parameters $p = 1.5$, 2.5, 3.5 and 4.5. Tapers that resist spectral leakage better (higher p) sample the ends of the time series less, suggesting that RF-analysis windows should start in the pre-event noise, not at the P-onset time. Right panel: composite frequency-sensitivity kernels $\mathcal{K}(f)$, summed over three eigentapers, for the different choices of time-bandwidth parameter p. For an identical data-window duration and number of tapers, higher p implies sensitivity with a wider central lobe (local-frequency averaging), and lower sidelobes (distant frequency leakage).
Figure S2. Left panel: composite frequency-sensitivity kernels $\mathcal{K}(f)$, summed over three eigentapers, for the different choices of time-series duration T, computed for time-bandwidth product $p = 2.5$. Longer time-series duration implies a narrower central lobe in the kernel. Right panel: Multiple-taper correlation (MTC) receiver functions for station RAYN of the Global Seismographic Network (GSN), computed for time windows of varying durations. An LQT rotation has been applied to the data before processing. The frequency cutoff $f_c = 2.0$ Hz. We filter the RFs to avoid bandedge ringing via a squared-cosine taper in the frequency domain, with half-amplitude at 1.0 Hz. We compute a weighted average of frequency-domain RFs from 164 seismic events in the $50^\circ-110^\circ$ back-azimuth sector, with epicentral distances $40^\circ < \Delta < 90^\circ$ halfwidth (http://gji.oxfordjournals.org/lookup/suppl/doi:10.1093/gji/ggw291/-/DC1).

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